

FOUR-LAYER DIAGNOSTIC-AWARE TELE-RADIOLOGY ARCHITECTURE (DARDO + CSA + 5G URLLC)

Mobile Stroke Unit (MSU) Scanner → Dual-Slice 5G Cellular RAN → Hospital Edge AI Safety Audit

TRANSMITTER: MOBILE STROKE UNIT (MSU)

LAYER 1: TASK PROFILER (CSA) — Lesion Prior

- **Raw Data:** Multi-coil raw k -space $\mathbf{Y} \in \mathbb{C}^{C \times K}$
- **Task Prior:** Spectrum $W(k_x, k_y) = |\mathcal{F}\{w_{CHO}\}|^2$

Safety Target: Detectability $d'_{CHO} \geq 1.5$

Prior $\hat{W}(k_x, k_y)$

LAYER 2: DIAGNOSTIC COMPRESSION (DARDO)

- **Diagnostic Grad:** $\gamma_k = \sum_y W(k, y) \|\mathcal{F}S_k\|_2$
- **Greedy Split:** Sort in $< 2\text{ms} \Rightarrow K_{\text{diag}}$ vs. K_{bulk}

Safety Probe: 128B Token θ in Header

URLLC Flow

$K_{\text{diag}} + \theta$

eMBB Flow

K_{bulk}

5G / B5G CELLULAR RADIO ACCESS NETWORK

LAYER 3: 5G URLLC SLICE (MISSION-CRITICAL)

- **Payload:** $K_{\text{diag}} = \{k \mid \gamma_k \geq \tau\} + \text{Token } \theta$
- **3GPP QoS:** QCI 1/65 (BLER $< 10^{-5}$, Latency $< 10\text{ms}$)

Channel: Dedicated BWP (Zero Burst Drop)

LAYER 3: 5G eMBB SLICE (BULK ANATOMY)

- **Bulk Payload:** $K_{\text{bulk}} = \{k \mid \gamma_k < \tau\}$ (Skull Context)
- **Policy:** Dynamic DASH streaming (Loss-tolerant)

Throughput: Best-Effort (Absorbs Congestion)

URLLC Flow

$K_{\text{diag}} + \theta$

eMBB Flow

K_{bulk}

RECEIVER: HOSPITAL EDGE SERVER

EDGE AI RECONSTRUCTION (VarNet)

- **Stream Fusion:** Fuses $\mathbf{Y}_{\text{URLLC}} \cup \mathbf{Y}_{\text{eMBB}}$
- **Deep Recon:** $\hat{\mathbf{X}} = \mathcal{R}_\theta(\mathbf{Y})$ in 1.8 s on Edge GPU

Residual: Passed $\|\hat{\mathbf{A}}\hat{\mathbf{X}} - \mathbf{Y}\| < \sigma_\eta$ (Null-Space Blind!)

Reconstructed $\hat{\mathbf{X}} + \theta$

LAYER 4: SEMANTIC AUDIT & DECISION GATE

✓ $z \geq 1.5$ (AUDIT PASSED: STROKE INTACT)

Score $z_{\text{recon}} \geq 1.5 \rightarrow$ Safe, Released to PACS & Clinician

✗ $z < 1.5$ (AUDIT FAILED: STROKE ERASED!)

Silent Erasure Alert $\Rightarrow$ Trigger Targeted S-NACK Recovery

Targeted S-NACK
(Resend Missing Δk)

CLOSED-LOOP S-NACK RETRANSMISSION PROTOCOL

Smart Recovery: Requests only missing high- γ_k bands ($\Delta k \approx 1.2$ MB, $\Delta t \approx 15$ ms) — Avoids 200 MB full retransmission

Audit Failed
(Stroke Erased)